

PCA & Eigenvectors

1. Matrix $\times$ Vector

Multiplying a matrix by a vector produces a **new** vector, generally with a different direction and length.

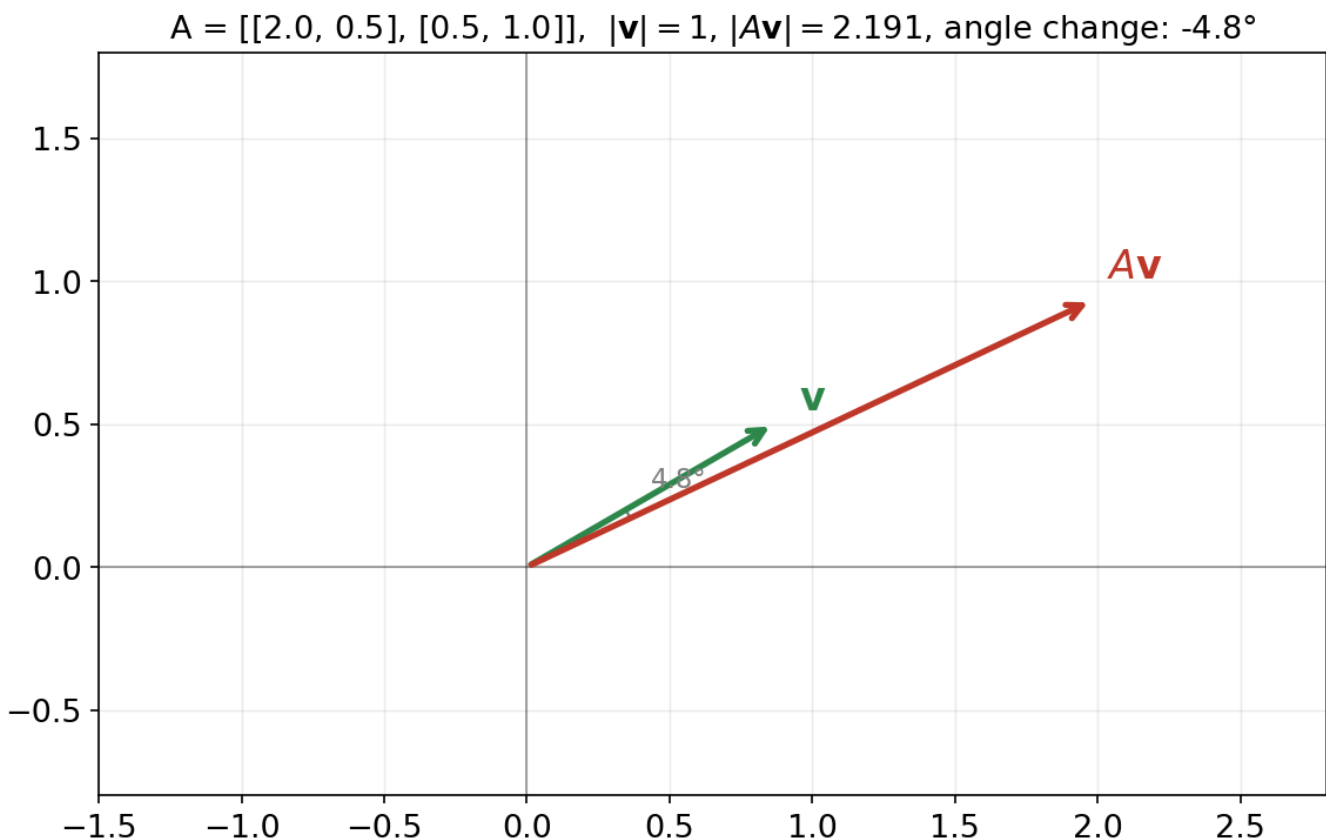
Given: Matrix $A = \begin{bmatrix} 2.0 & 0.5 \\ 0.5 & 1.0 \end{bmatrix}$, input vector $\mathbf{v}$ at $\theta = 30^\circ$ (unit length).

$$\mathbf{v} = \begin{bmatrix} \cos 30^\circ \\ \sin 30^\circ \end{bmatrix} = \begin{bmatrix} 0.8660 \\ 0.5000 \end{bmatrix}$$

$$A\mathbf{v} = \begin{bmatrix} 2.0 & 0.5 \\ 0.5 & 1.0 \end{bmatrix} \begin{bmatrix} 0.8660 \\ 0.5000 \end{bmatrix} = \begin{bmatrix} (2.0)(0.8660) + (0.5)(0.5000) \\ (0.5)(0.8660) + (1.0)(0.5000) \end{bmatrix} = \begin{bmatrix} 1.9821 \\ 0.9330 \end{bmatrix}$$

- $|\mathbf{v}| = 1$
- $|A\mathbf{v}| = 2.1907$
- Angle change: -4.8°

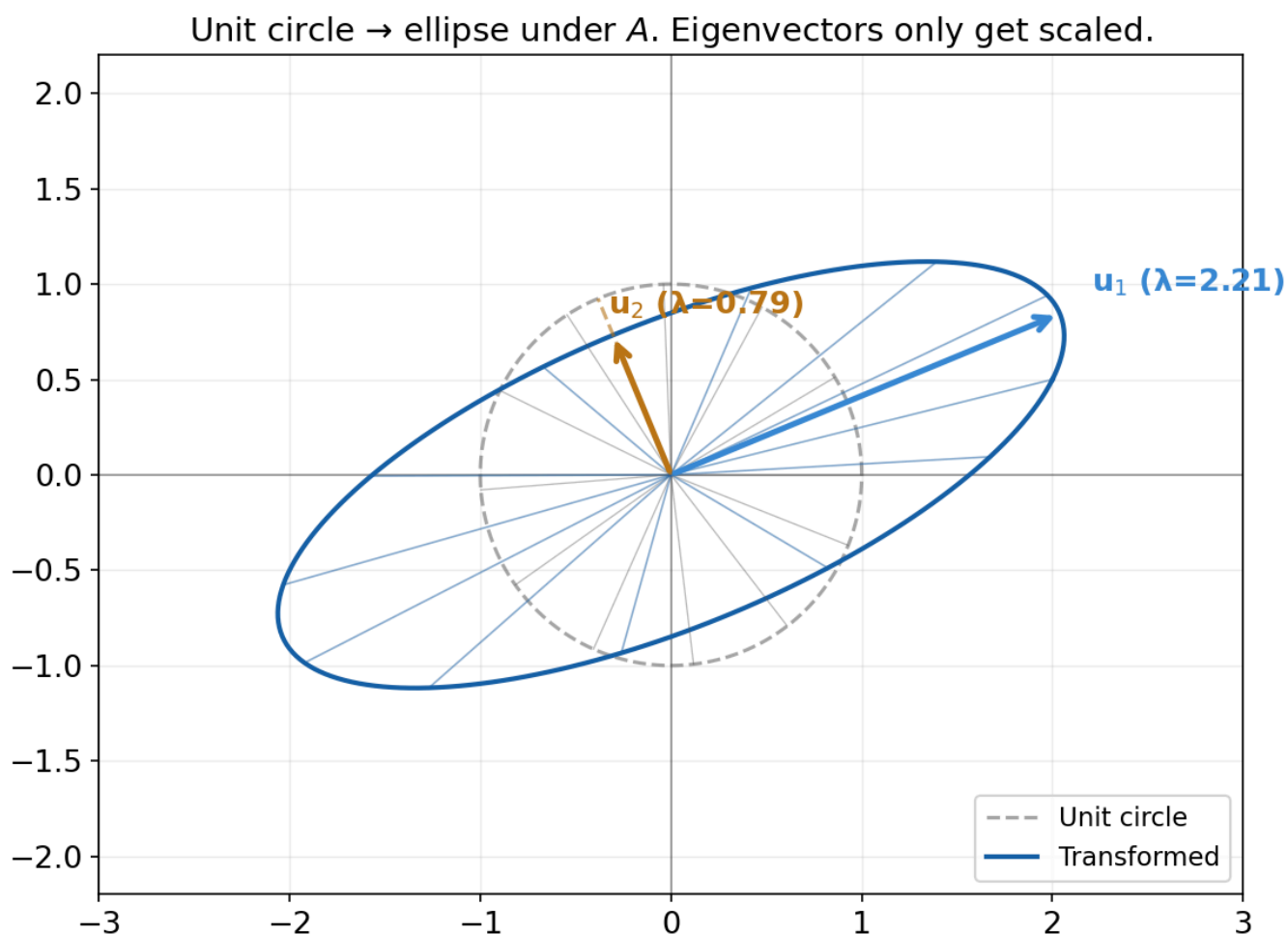
The output vector is longer and slightly rotated.



2. Geometric Intuition

When a matrix transforms the unit circle, most vectors change both direction and length. But **eigenvectors only get scaled**, they stay on their original line.

Using $A = \begin{bmatrix} 2.0 & 0.5 \\ 0.5 & 1.0 \end{bmatrix}$:



- The **unit circle** (dashed) maps to an **ellipse** (blue)
- u_1 (blue arrow) only stretches by $\lambda_1 = 2.2071$, no rotation
- u_2 (amber arrow) only stretches by $\lambda_2 = 0.7929$, no rotation
- All other vectors rotate and stretch

3. Eigenvalue Equation

Step 1: Definition

$$A\mathbf{u} = \lambda\mathbf{u} \quad (\mathbf{u} \neq \mathbf{0})$$

The matrix A only **scales** $\mathbf{u}$, no rotation. λ is that scale factor.

Step 2: Rearrange

$$A\mathbf{u} - \lambda\mathbf{u} = \mathbf{0} \Rightarrow (A - \lambda I)\mathbf{u} = \mathbf{0}$$

For $\mathbf{u} \neq \mathbf{0}$, the matrix $(A - \lambda I)$ must be singular:

$$\det(A - \lambda I) = 0 \quad \leftarrow \text{characteristic equation}$$

Step 3: Expand the determinant

Using $A = \begin{bmatrix} 2 & 0.5 \\ 0.5 & 1 \end{bmatrix}$:

$$\det \begin{pmatrix} 2 - \lambda & 0.5 \\ 0.5 & 1 - \lambda \end{pmatrix} = (2 - \lambda)(1 - \lambda) - (0.5)(0.5) = 0$$

$$\lambda^2 - 3\lambda + 1.75 = 0$$

Discriminant: $\Delta = 3^2 - 4(1.75) = 2.0 \Rightarrow \text{real eigenvalues}$

$$\lambda = \frac{3 \pm \sqrt{2}}{2}$$

$$\lambda_1 = 2.20711, \quad \lambda_2 = 0.79289$$

Step 4: Recover eigenvectors

For each λ_i , substitute back into $(A - \lambda_i I)\mathbf{u} = \mathbf{0}$. Since $\det(A - \lambda_i I) = 0$, the system is underdetermined (the two rows are linearly dependent). We set $u_2 = 1$, solve for u_1 , then normalise.

For $\lambda_1 = 2.2071$:

$$\begin{bmatrix} -0.207 & 0.500 \\ 0.500 & -1.207 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Row 1: $-0.207 \cdot u_1 + 0.500 \cdot u_2 = 0$

Set $u_2 = 1 \Rightarrow u_1 = \frac{-(0.500)}{-0.207} = 2.4142$

Unnormalized: $\mathbf{u}' = [2.4142, 1]$

$$\|\mathbf{u}'\| = \sqrt{2.4142^2 + 1^2} = \sqrt{5.8284 + 1} = 2.6131$$

$$\mathbf{u}_1 = \frac{1}{2.6131} \begin{bmatrix} 2.4142 \\ 1 \end{bmatrix} = \begin{bmatrix} 0.9239 \\ 0.3827 \end{bmatrix}$$

For $\lambda_2 = 0.7929$:

$$\begin{bmatrix} 1.207 & 0.500 \\ 0.500 & 0.207 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Row 1: $1.207 \cdot u_1 + 0.500 \cdot u_2 = 0$

Set $u_2 = 1 \Rightarrow u_1 = \frac{-(0.500)}{1.207} = -0.4142$

Unnormalized: $\mathbf{u}' = [-0.4142, 1]$

$$\|\mathbf{u}'\| = \sqrt{0.1716 + 1} = 1.0824$$

$$\mathbf{u}_2 = \frac{1}{1.0824} \begin{bmatrix} -0.4142 \\ 1 \end{bmatrix} = \begin{bmatrix} -0.3827 \\ 0.9239 \end{bmatrix}$$

4. Covariance Matrix

The dataset

We use $N = 25$ data points in $\mathbb{R}^2$.

i	x_1	x_2	----	i	x_1	x_2	----	i	x_1	x_2
1	2.1	2.2	----	10	1.7	0.2	----	19	0.1	0.4
2	1.5	0.3	----	11	3.5	3.2	----	20	-2.4	-1.4
3	3.2	2.1	----	12	-0.8	0.2	----	21	3.0	1.6
4	-0.5	-1.6	----	13	2.3	1.0	----	22	1.8	2.0
5	1.0	1.2	----	14	-2.1	-2.6	----	23	-1.0	-1.8
6	-1.2	-0.8	----	15	0.7	1.0	----	24	0.5	-0.6
7	2.8	1.0	----	16	1.2	-0.2	----	25	-0.3	0.6
8	0.3	-1.0	----	17	-1.5	-1.8	----			
9	-1.8	-2.3	----	18	2.6	2.6	----			

Mean:

$$\boldsymbol{\mu} = [0.6680, 0.2200]$$

Covariance matrix (computed from centered data $\tilde{X} = (X - \boldsymbol{\mu})$):

$$\Sigma = \frac{1}{N} \tilde{X}^\top \tilde{X} = \begin{bmatrix} 3.0150 & 2.3958 \\ 2.3958 & 2.4728 \end{bmatrix}$$

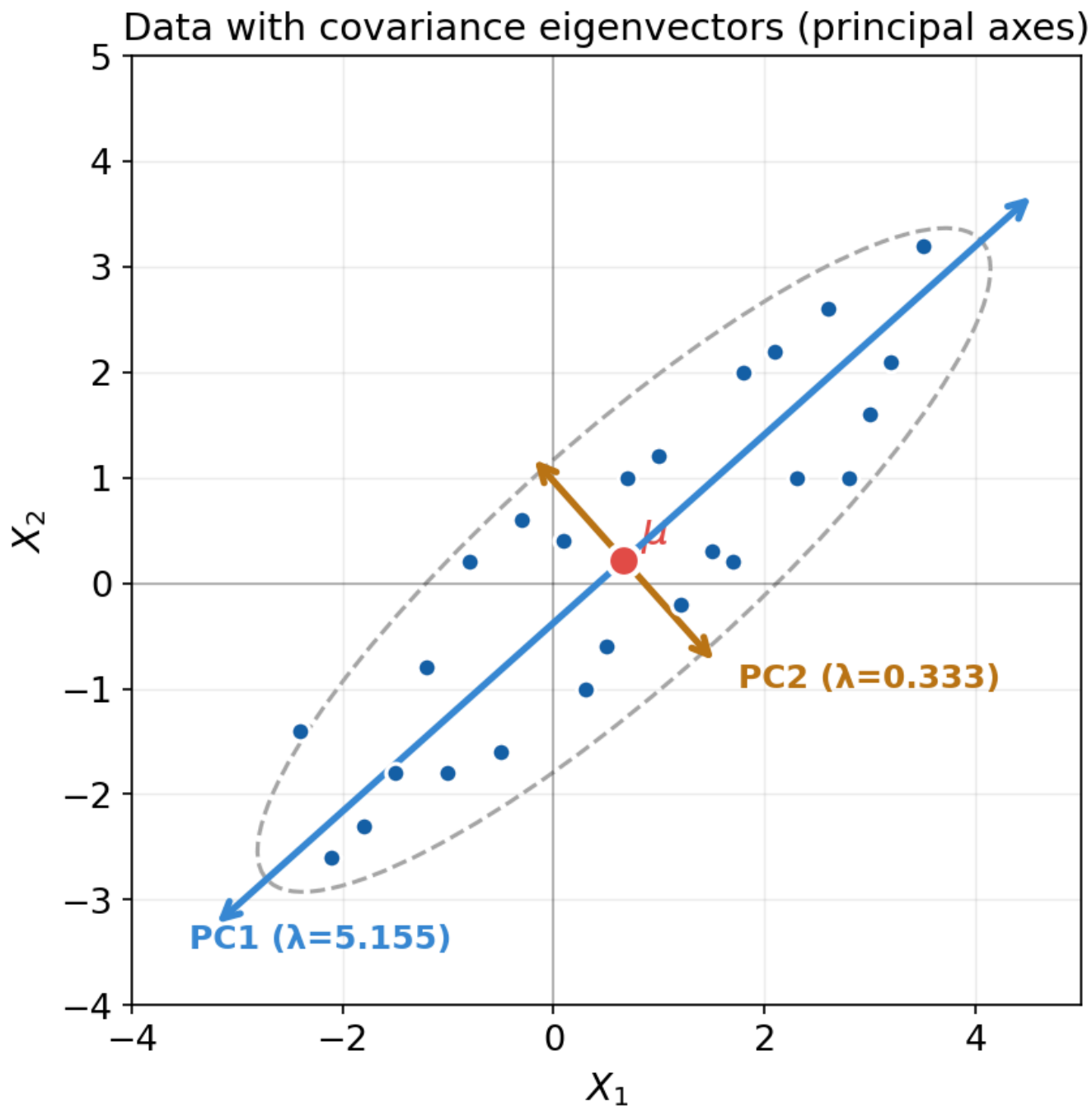
- Diagonal entries = variances of each feature: $\text{Var}(X_1) = 3.0150$, $\text{Var}(X_2) = 2.4728$
- Off-diagonal = covariance: $\text{Cov}(X_1, X_2) = 2.3958$ (strongly positively correlated)

Eigendecomposition: $\Sigma = U\Lambda U^\top$

$$\lambda_1 = 5.1550, \quad \mathbf{u}_1 = [0.7458, 0.6662] \quad \rightarrow \text{max variance direction}$$

$$\lambda_2 = 0.3328, \quad \mathbf{u}_2 = [0.6662, -0.7458] \quad \rightarrow \text{orthogonal to } \mathbf{u}_1$$

Variance explained: PC1 = 93.9%, PC2 = 6.1%



Why the covariance matrix?

Project all data onto a unit vector $\mathbf{w}$: each projected value is $z_i = \mathbf{w}^\top \tilde{\mathbf{x}}_i$.

$$\text{Var}(z) = \frac{1}{N} \sum_{i=1}^N (\mathbf{w}^\top \tilde{\mathbf{x}}_i)^2 = \mathbf{w}^\top \underbrace{\left(\frac{1}{N} \sum_i \tilde{\mathbf{x}}_i \tilde{\mathbf{x}}_i^\top \right)}_{\Sigma} \mathbf{w} = \mathbf{w}^\top \Sigma \mathbf{w}$$

The variance of the projection onto *any* direction $\mathbf{w}$ equals $\mathbf{w}^\top \Sigma \mathbf{w}$. The covariance matrix Σ completely encodes how variance is distributed across all directions. Decomposing it reveals which directions carry the most variance.

5. Why PCA Works

Variance maximization: constrained optimization

PCA asks: **which unit vector $\mathbf{w}$ maximises $\mathbf{w}^\top \Sigma \mathbf{w}$?**

$$\max_{\mathbf{w}} \mathbf{w}^\top \Sigma \mathbf{w} \quad \text{subject to} \quad \mathbf{w}^\top \mathbf{w} = 1$$

Introduce a Lagrange multiplier λ to encode the constraint:

$$\mathcal{L}(\mathbf{w}, \lambda) = \mathbf{w}^\top \Sigma \mathbf{w} - \lambda(\mathbf{w}^\top \mathbf{w} - 1)$$

Take the derivative and set to zero:

$$\frac{\partial \mathcal{L}}{\partial \mathbf{w}} = 2\Sigma \mathbf{w} - 2\lambda \mathbf{w} = \mathbf{0}$$

$$\boxed{\Sigma \mathbf{w} = \lambda \mathbf{w}}$$

This is exactly the **eigenvalue equation**. The optimal direction must be an eigenvector of Σ .

Variance of projection = eigenvalue

Substitute $\Sigma \mathbf{w} = \lambda \mathbf{w}$ back:

$$\text{Var}(z) = \mathbf{w}^\top \Sigma \mathbf{w} = \mathbf{w}^\top (\lambda \mathbf{w}) = \lambda \underbrace{\mathbf{w}^\top \mathbf{w}}_{=1} = \lambda$$

The variance along an eigenvector **equals its eigenvalue**. To maximise variance, choose the eigenvector with the **largest** eigenvalue, that is **PC1**.

The second PC maximises variance subject to $\mathbf{w}^\top \mathbf{w} = 1$ *and* orthogonality to PC1. Solving gives the eigenvector with the second largest eigenvalue. And so on.

Why eigenvectors give uncorrelated components

Let $\mathbf{u}_i$ and $\mathbf{u}_j$ be eigenvectors of Σ with eigenvalues λ_i, λ_j . Their projections are $z^{(i)} = \mathbf{u}_i^\top \tilde{\mathbf{x}}$ and $z^{(j)} = \mathbf{u}_j^\top \tilde{\mathbf{x}}$.

$$\text{Cov}(z^{(i)}, z^{(j)}) = \mathbf{u}_i^\top \Sigma \mathbf{u}_j = \mathbf{u}_i^\top (\lambda_j \mathbf{u}_j) = \lambda_j \underbrace{\mathbf{u}_i^\top \mathbf{u}_j}_{=0 \text{ (orthogonal)}} = 0$$

Since Σ is symmetric, its eigenvectors are orthogonal. Therefore the projected components are **uncorrelated by construction**.

Change of basis: diagonalising the covariance

Assemble all eigenvectors as columns into U and eigenvalues into Λ :

$$\Sigma = U\Lambda U^\top$$

Transform data into eigenvector coordinates: $Z = \tilde{X}U$. The covariance of Z is:

$$\Sigma_Z = \frac{1}{N} Z^\top Z = U^\top \Sigma U = U^\top (U\Lambda U^\top) U = \Lambda$$

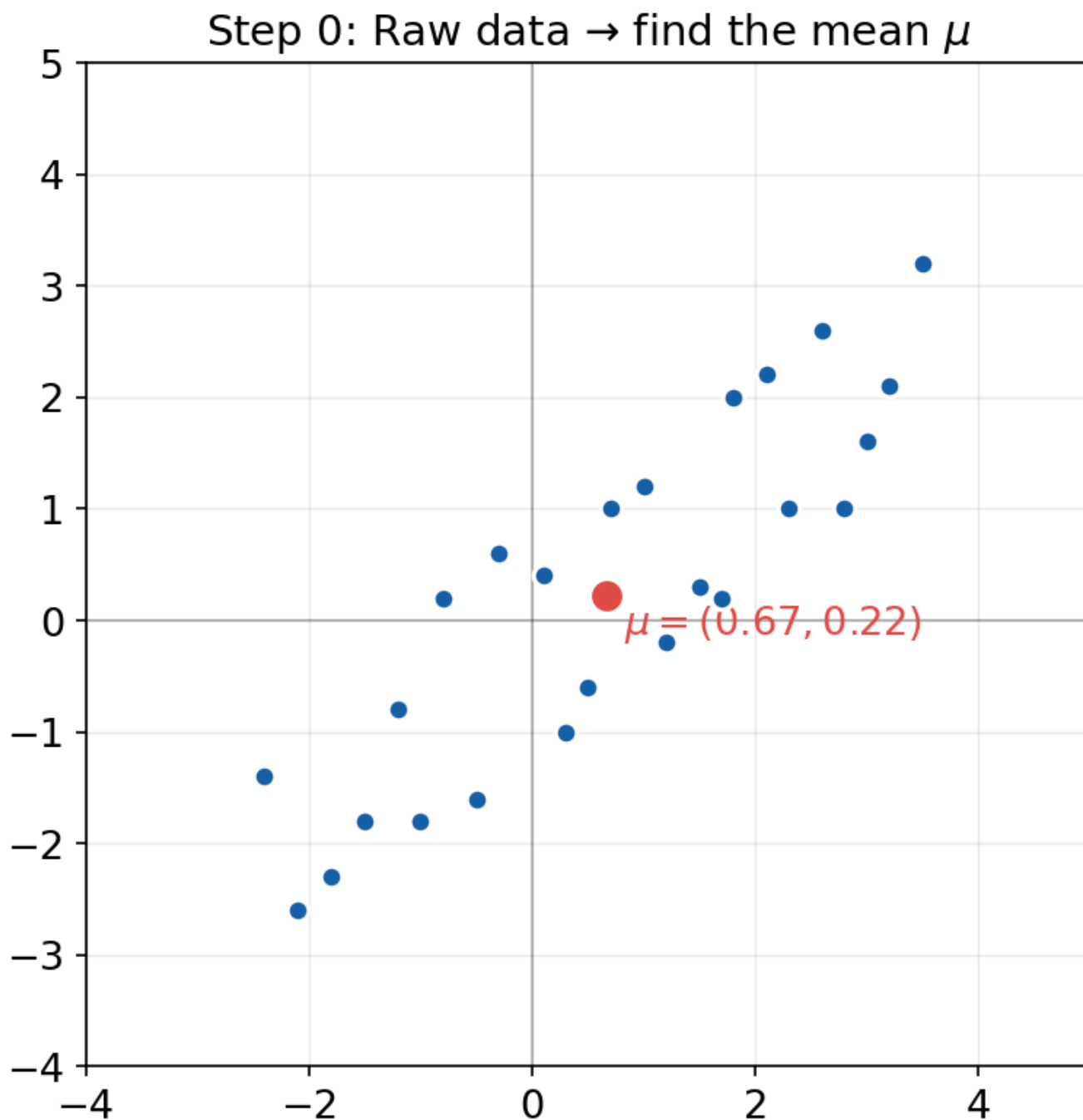
The covariance matrix of the projected data is **diagonal**, i.e, all off-diagonal covariances are zero. PCA finds the coordinate system where the data's covariance structure is completely diagonal: variance only, no covariance.

6. PCA Algorithm

Working with the same $N = 25$ dataset.

Step 0: Raw data

Find the mean: $\mu = [0.6680, 0.2200]$.

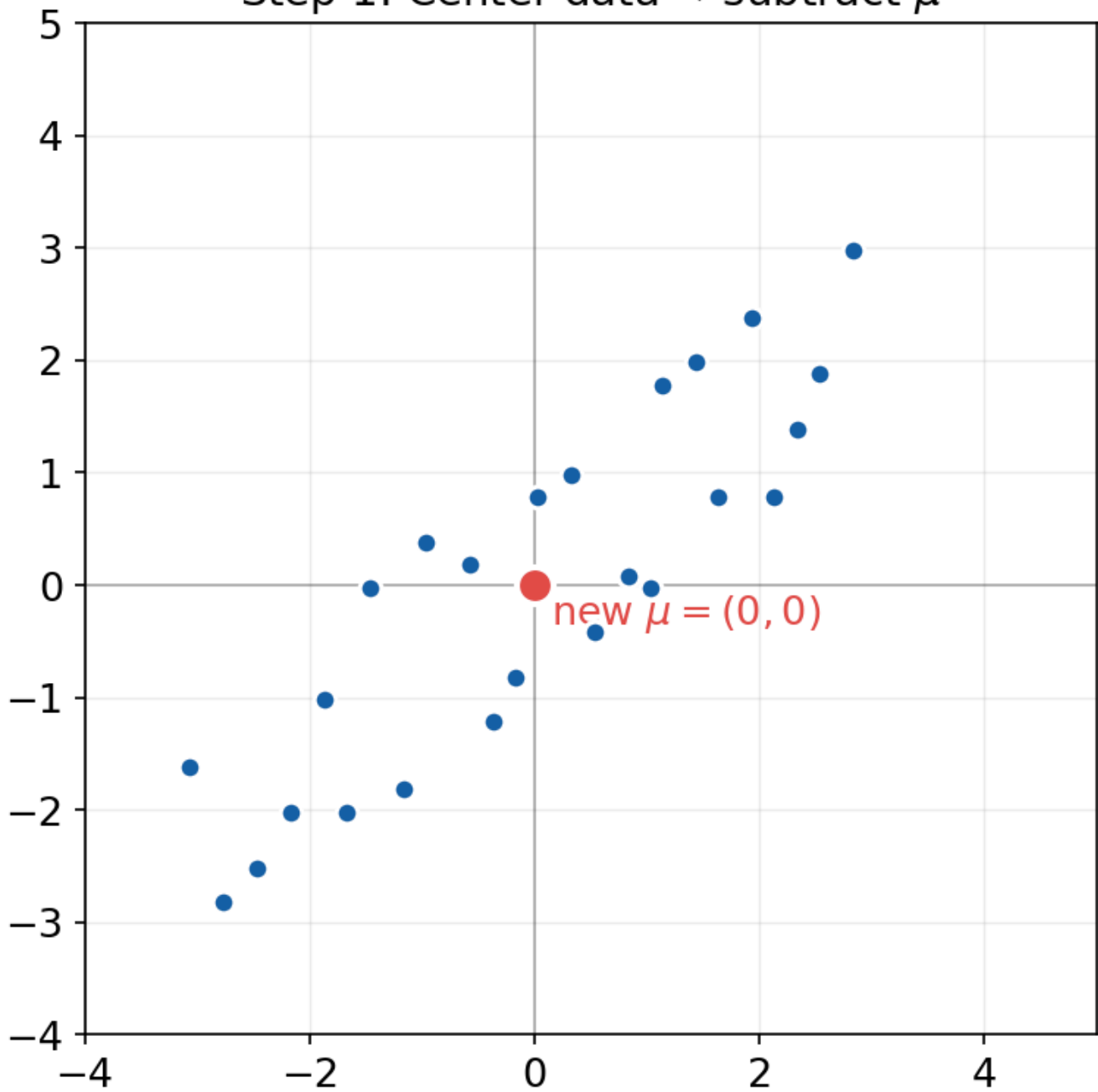


Step 1: Center the data

Subtract μ from every point: $\tilde{x}_i = x_i - \mu$

The centered data $\tilde{X}$ has zero mean in every column.

Step 1: Center data $\rightarrow$ subtract μ

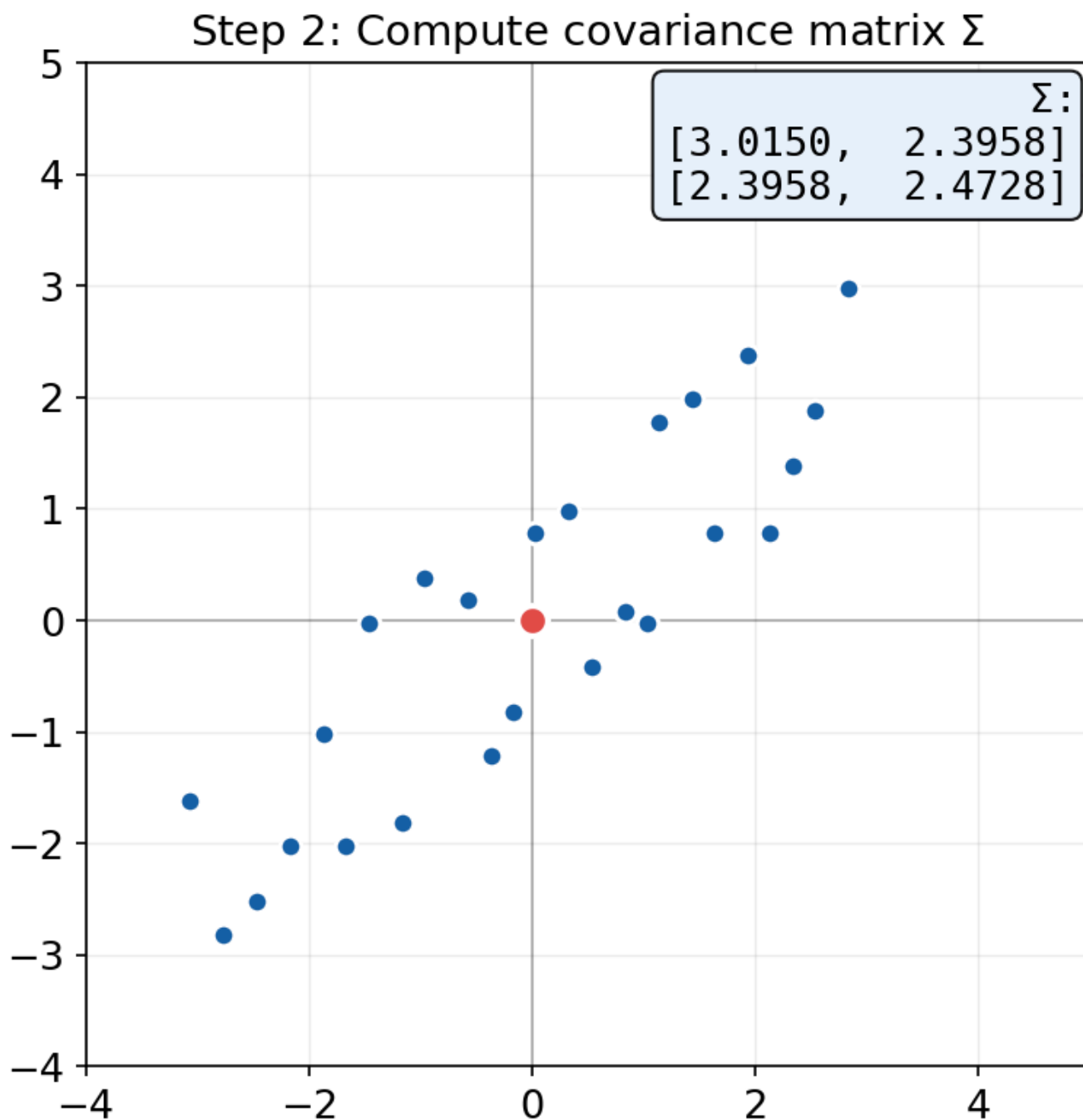


Step 2: Compute the covariance matrix

$$\Sigma = \frac{1}{N} \tilde{X}^\top \tilde{X} = \begin{bmatrix} 3.0150 & 2.3958 \\ 2.3958 & 2.4728 \end{bmatrix}$$

Diagonal entries = variances of each feature. Off-diagonal = covariance (how correlated they

are).



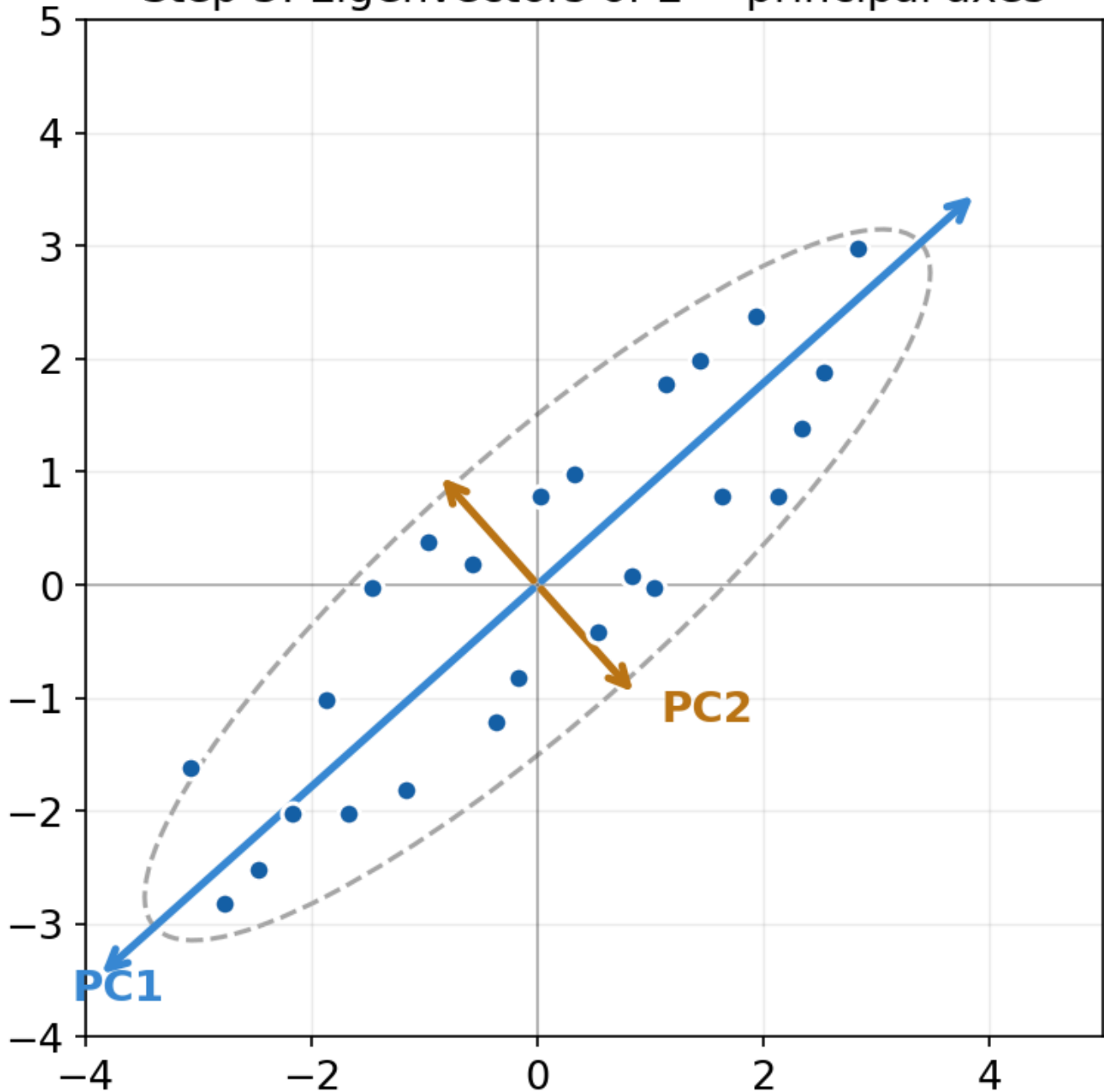
Step 3: Eigenvectors of Σ

Eigenvectors point along the principal axes of the data ellipse. The larger eigenvalue $\rightarrow$ direction of maximum spread.

$$\lambda_1 = 5.1550, \quad \mathbf{u}_1 = [0.7458, 0.6662]$$

$$\lambda_2 = 0.3328, \quad \mathbf{u}_2 = [0.6662, -0.7458]$$

Step 3: Eigenvectors of $\Sigma \rightarrow$ principal axes



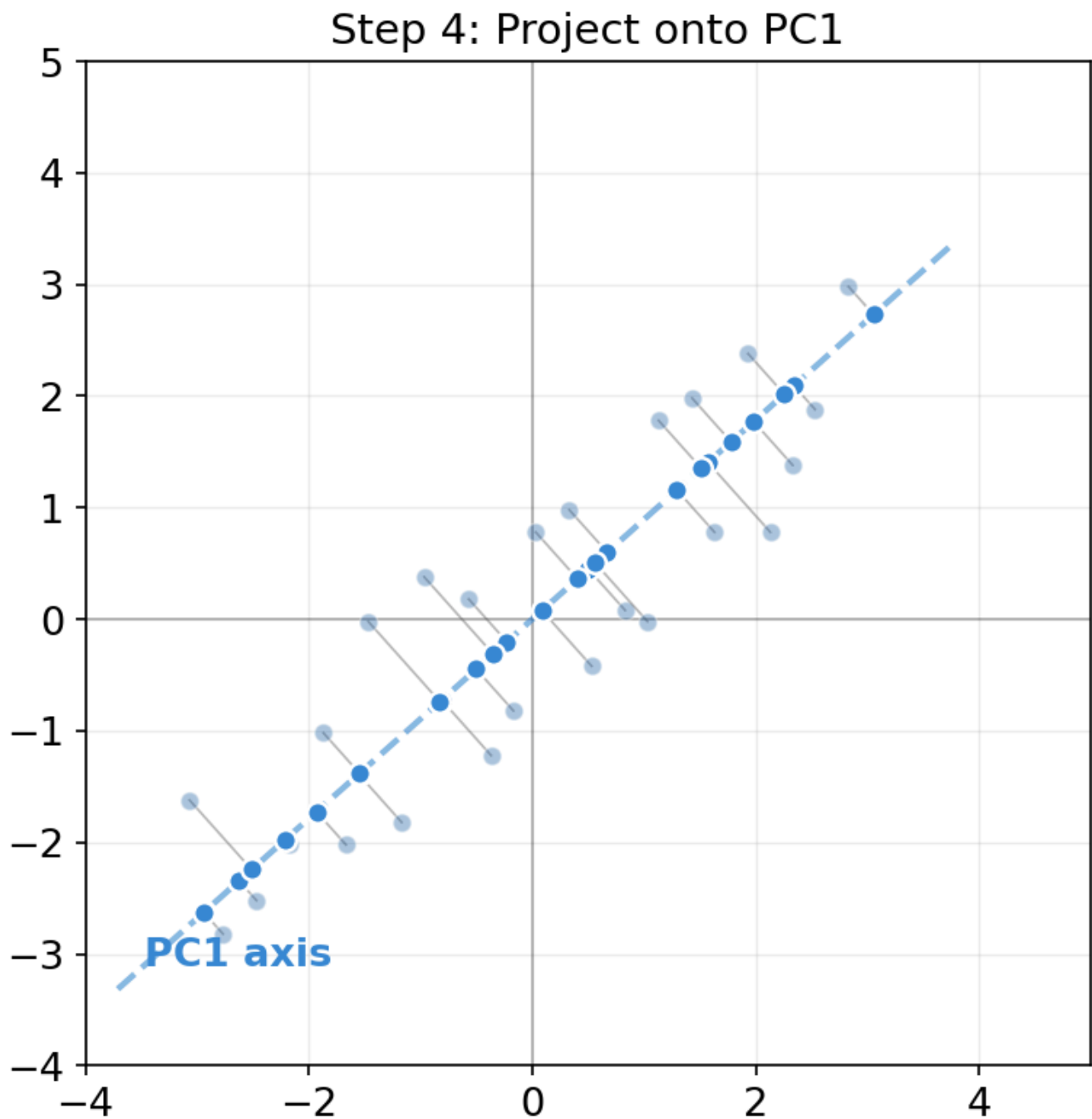
Step 4: Project the data

Project every centered point onto the eigenvectors: $Z = \tilde{X}U$ where $U = [\mathbf{u}_1 \mid \mathbf{u}_2]$.

$$z_i^{(1)} = \tilde{\mathbf{x}}_i \cdot \mathbf{u}_1, \quad z_i^{(2)} = \tilde{\mathbf{x}}_i \cdot \mathbf{u}_2$$

Each point now has coordinates in the PC basis. The dashed lines show the projection onto

PC1.



Step 5: Verify and decide

Verify: The variance of the projected data along each PC should equal the corresponding eigenvalue.

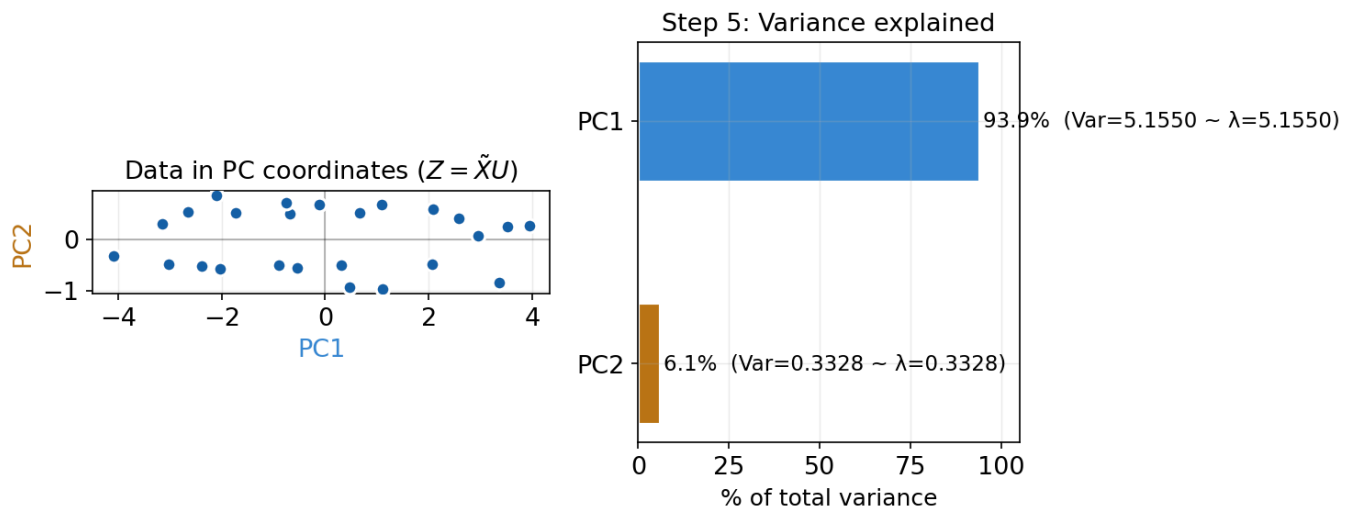
$$\text{Var}(Z_1) = \frac{1}{N} \sum z_i^{(1)2} = 5.1550 = \lambda_1 \checkmark$$

$$\text{Var}(Z_2) = \frac{1}{N} \sum z_i^{(2)2} = 0.3328 = \lambda_2 \checkmark$$

Decide: PC1 explains **93.9%** of total variance. Keeping only PC1 reduces from $p = 2$ to $d = 1$

dimensions while retaining most information.

$$Z_{\text{reduced}} = \tilde{X} \cdot \mathbf{u}_1 \quad (N \times 1 \text{ vector})$$



7. PCA Limitations

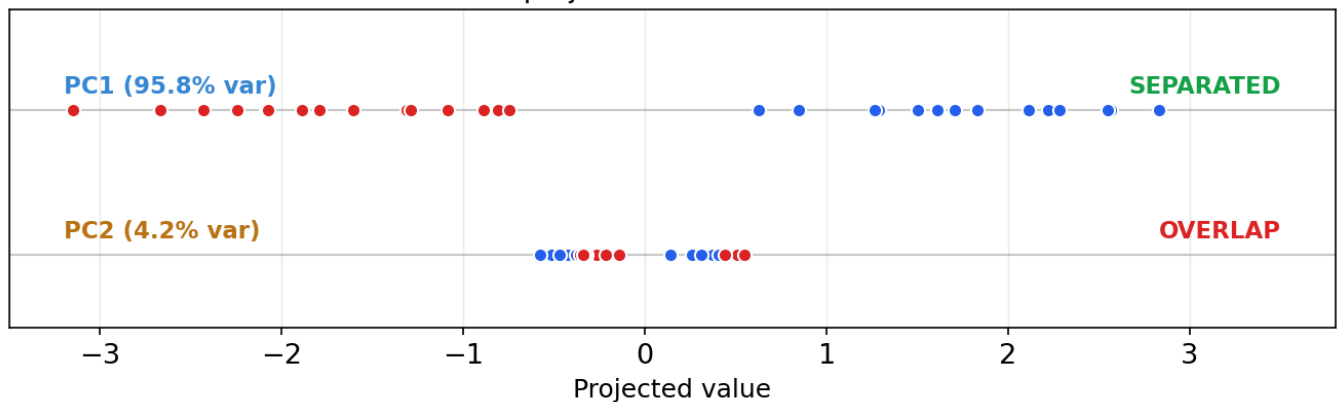
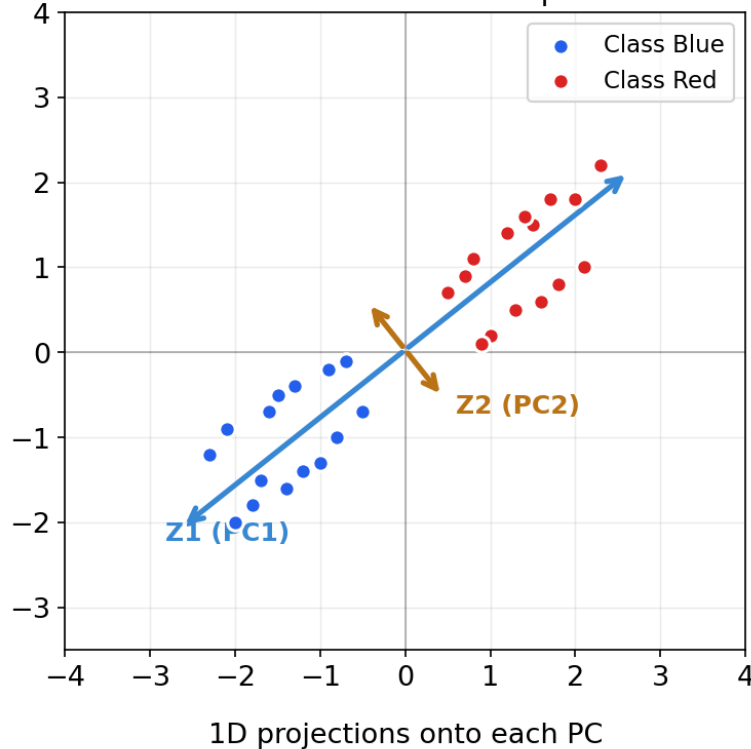
PCA is **unsupervised**. It maximises variance, not class separation. The direction of largest variance may not be the most discriminative for classification.

Case A: PCA works well

Z_1 (PC1) captures the most variance **and** separates the classes. Projecting from $p = 2$ to $d = 1$ via PCA keeps the discriminative information.

PCA chooses Z_1 ✓ Classes are separated along Z_1

PCA Limitations → Case A: PC1 separates classes



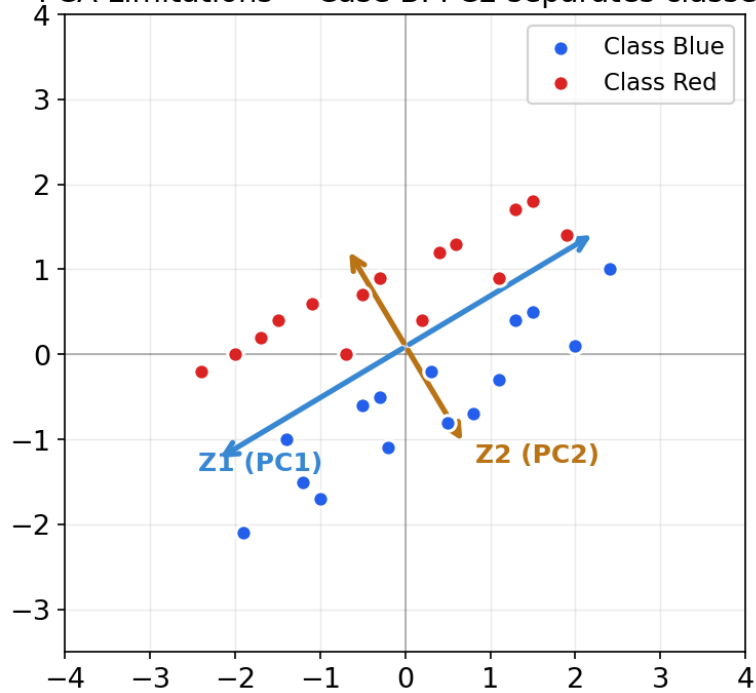
Case B: PCA discards discriminative information

Z_1 (PC1) captures most variance but the classes **overlap** when projected onto it. Z_2 (PC2) captures little variance but **perfectly separates** the classes.

PCA would keep Z_1 and discard Z_2 , losing the classification signal!

PCA chooses $Z_1 \propto$ but Z_2 is discriminative

PCA Limitations → Case B: PC2 separates classes



1D projections onto each PC

